Overlooked model uncertainties may misinform forest management strategies

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Supplementary methods

Calibration of process-based models

The process-based model we used, PHENOFIT, was developed for several European tree species and validated by comparing species distribution to the model outputs (Saltré et al., 2013; Duputié et al., 2015; Gauzere et al., 2020; Van der Meersch et al., 2025). Some parameters were directly measured or found in the literature (e.g. frost hardiness parameters). Phenology-related parameters were inferred using phenological observations across Europe – provided through the TEMPO data portal (`data.pheno.fr`), the PEP725 database (`pep725.eu`) and the PhenoClim citizen science program (`https://spot.creamontblanc.org/en/programs/phenoclim`).

Species distribution data

The calibration of correlative and hybrid models requires species presence data across all Europe. The data we used come mainly from forest inventories conducted across most European countries (Mauri et al., 2017). We completed it with presence records from the Global Biodiversity Information Facility (GBIF, 2022), removing occurrences outside the natural ranges of the species as defined by Atlas Florae Europaeae (Jalas and Suominen, 2005) and EuroVegMap (Bohn et al., 2003). For the two mediterranean species included in this study (*Q. ilex* and *Q. pubescens*), we also used occurrences from the WOODIV database (Monnet et al., 2021). See Van der Meersch et al. (2025) for the details on the processing of occurrence data.

Calibration of correlative models

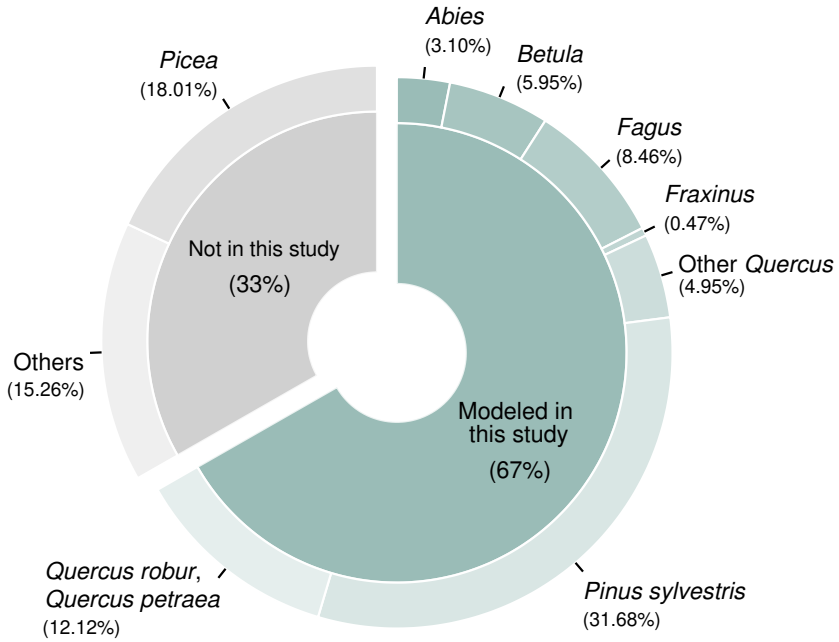
We used four different models: GLM, GAM, BRT and Random Forest. We applied the tuning recommendations of Valavi et al. (2022):
- for GAM, GLM and BRT , background points were down-weighted so that their total weight was equal to the total weight of the presence points
- for GLM, we allowed both linear and quadratic terms and applied a lasso regularization in order to avoid overfitting issues and improve model transferability. The model was tuned with the internal cross-validation feature of *cv.glmnet*
- for BRT, we estimated the optimal number of trees using the forward stepwise cross-validation approach implemend in the R package *dismo*
- for Random Forest, each tree was fitted with a bootstrap sample of presences and the same number of background points

Calibration of hybrid models

Hybrid models correspond to process-based models entirely calibrated using species distribution data, as correlative models. We use the procedure described in Van der Meersch and Chuine (2023), using the covariance matrix adaptation evolution strategy (CMA-ES), implemented on

the GenOuest multicore cluster (genouest.org). This robust algorithm allows to efficiently calibrate complex models (Hansen and Ostermeier, 2001). We calibrated each species parameter set 10 times, using five repetitions on two random subsets of presences and pseudo-absences (see Van der Meersch et al. (2025) for details). We thus obtain ten different hybrid model version for each species.

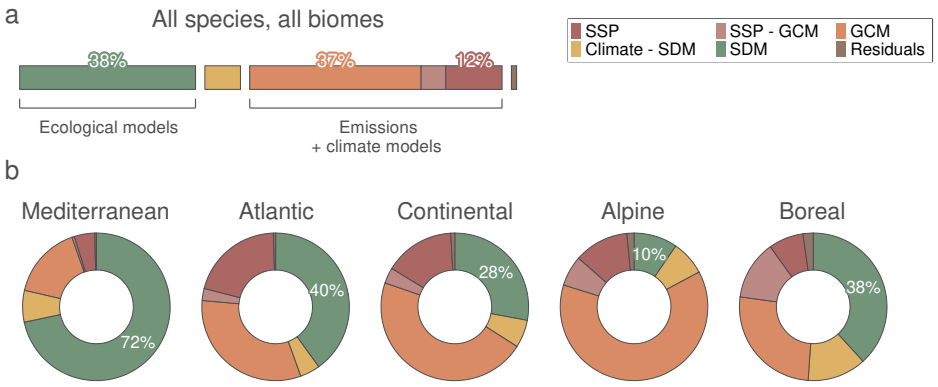
Supplementary figures



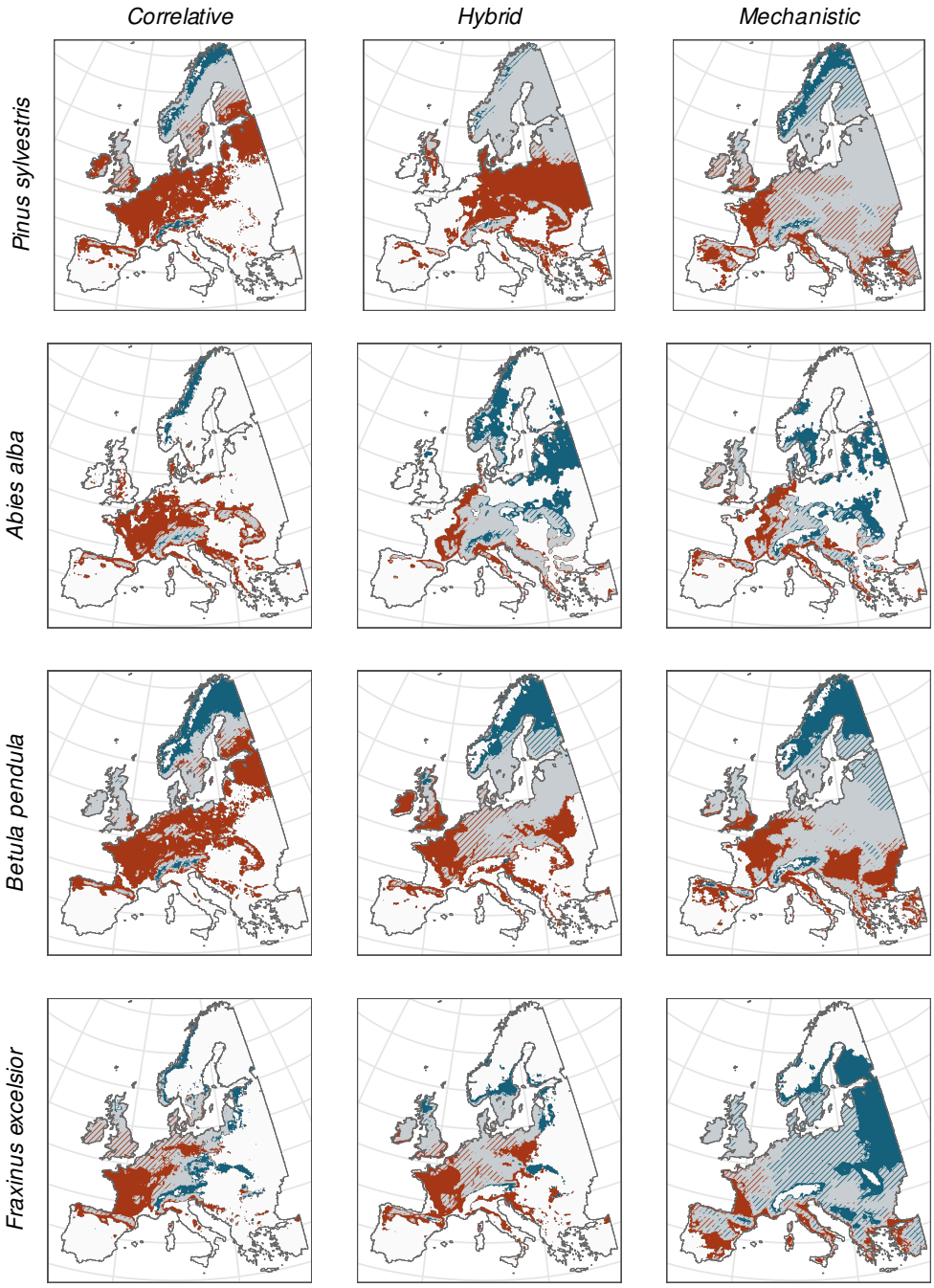
Supplementary Figure 1: Share of European forest area by dominant tree species or genus. Data at 1-km spatial resolution come from Brus et al. (2011). A grid cell is considered forested if more than 10% of its surface is covered by forest.

Supplementary Table 1: Projected change in habitat suitability and area loss for Fagus sylvatica within its historical distribution, according to SSP2-4.5 (2080-2100). Values represent the mean across model projections, with standard deviations in parentheses.

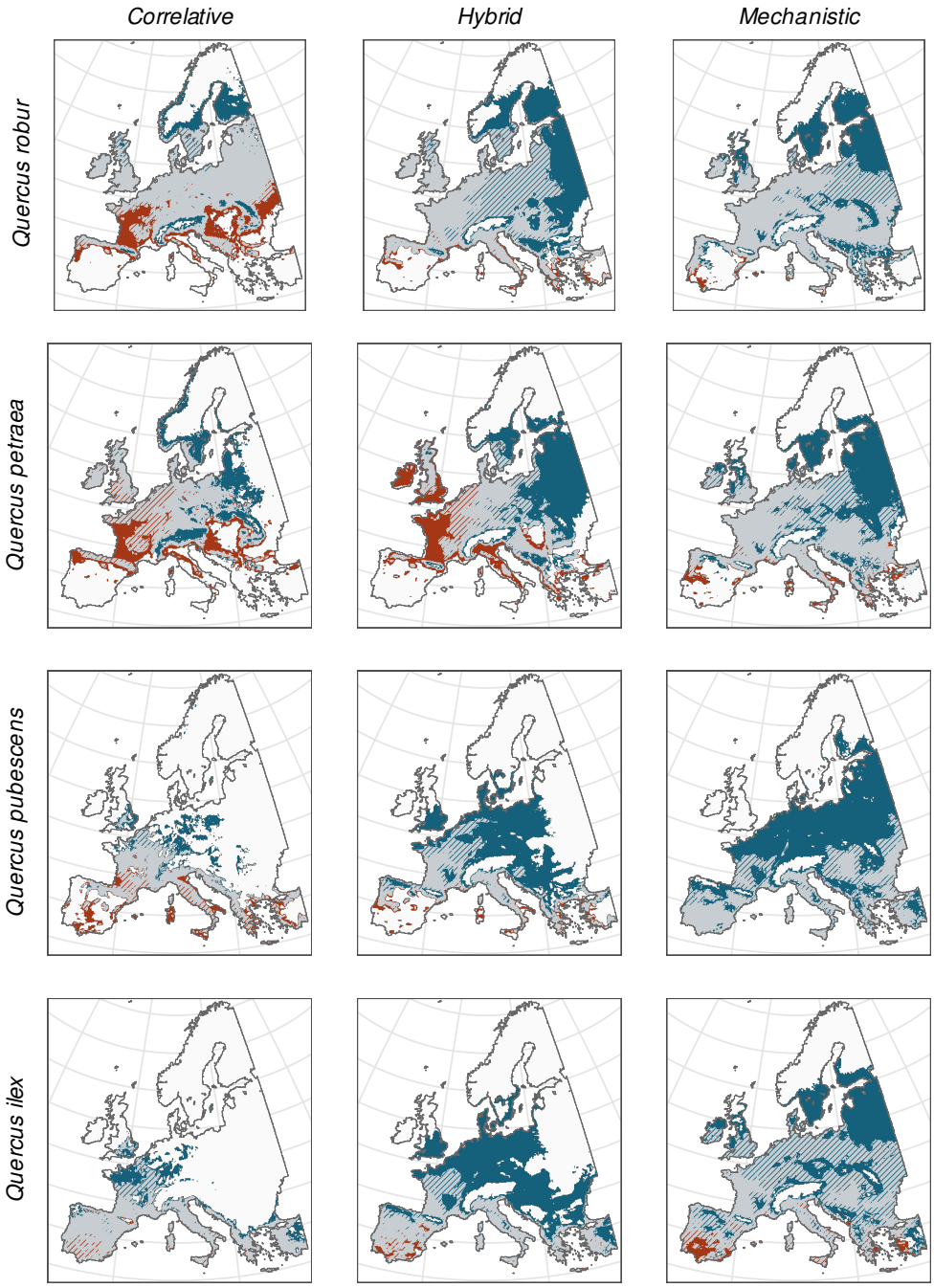
	Mechanistic		
	Calibration with distribution data		
	Correlative	Hybrid	Process-based
Change in suitability	-0.157 (0.0515)	-0.113 (0.106)	0.185 (0.0279)
	-0.135 (0.0821)		
		-0.0282 (0.170)	
Percentage loss of area	29.3% (13.3)	31.7% (16.6)	4.48% (2.38)
	30.5% (14.2)		
		21.8% (17.1)	



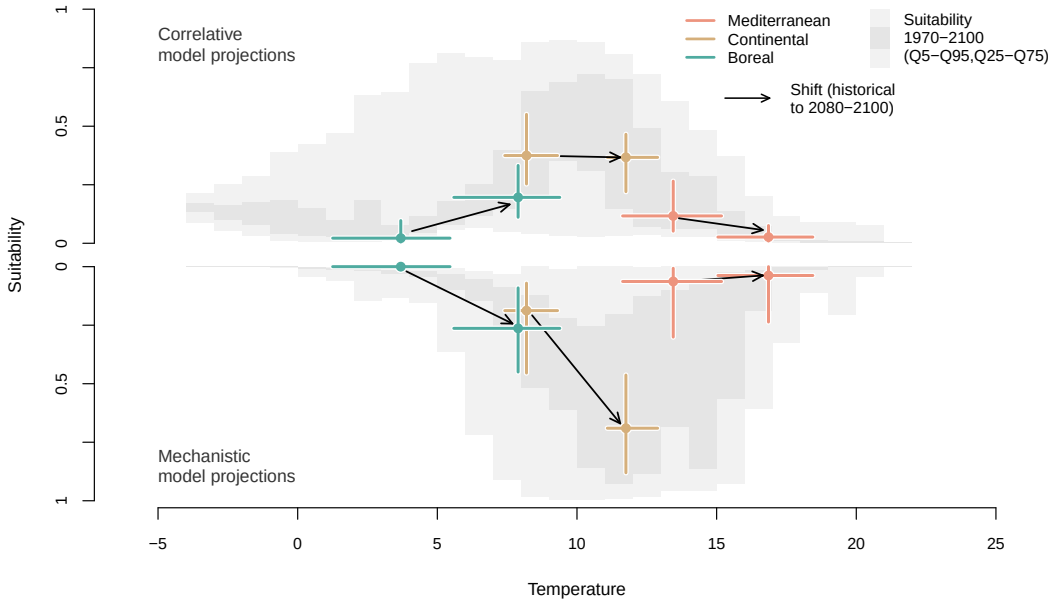
Supplementary Figure 2: Considering only correlative models leads to an underestimation of ecological model uncertainty. This figure illustrates the level of uncertainty associated with projected changes in suitability relative to the historical period (1970–2000), **(a)** across all species and biomes and **(b)** for each biome. Here, we considered that the ecological model uncertainty is only explained by the differences between correlative modeling approaches.



Supplementary Figure 3: Average change in projected distribution under scenario SSP2-4.5 (2080-2100 relative to 1970-2000), for three ecological modeling approaches (correlative, hybrid, mechanistic).



Supplementary Figure 4: Average change in projected distribution under scenario SSP2-4.5 (2080-2100 relative to 1970-2000), for three ecological modeling approaches (correlative, hybrid, mechanistic).



Supplementary Figure 5: Geographic patterns of projection uncertainties reflect the position of species' thermal niches. In the continental biome, climatic conditions are closer to modelled thermal optima, leading to greater uncertainty depending on whether species shift toward or away from their optimum. Grey shading represents species suitability within each 1°C bin over 1970-2100 (light grey: Q5-Q95; dark grey: Q25-Q75).

References

Bohn, U., Neuhäusl, R., Gollub, G., Hettwer, C., Neuhäuslová, Z., Raus, T., Schlüter, H., and Weber, H. (2003). Map of the natural vegetation of europe - scale 1:2500000.

Brus, D. J., Hengeveld, G. M., Walvoort, D. J. J., Goedhart, P. W., Heidema, A. H., Nabuurs, G. J., and Gunia, K. (2011). Statistical mapping of tree species over europe. *European Journal of Forest Research*, 131(1):145–157.

Duputié, A., Rutschmann, A., Ronce, O., and Chuine, I. (2015). Phenological plasticity will not help all species adapt to climate change. *Global Change Biology*, 21(8):3062–3073.

Gauzere, J., Teuf, B., Davi, H., Chevin, L.-M., Caignard, T., Leys, B., Delzon, S., Ronce, O., and Chuine, I. (2020). Where is the optimum? Predicting the variation of selection along climatic gradients and the adaptive value of plasticity. A case study on tree phenology. *Evolution Letters*, 4(2):109–123.

GBIF (2022). The global biodiversity information facility.

Hansen, N. and Ostermeier, A. (2001). Completely Derandomized Self-Adaptation in Evolution Strategies. *Evolutionary Computation*, 9(2):159–195.

Jalas, J. and Suominen, J. (1972–2005). *Atlas Florae Europaeae*. Committee for Mapping the Flora of Europe and Societas Biologica Fennica Vanamo, Helsinki, Finland.

Mauri, A., Strona, G., and San-Miguel-Ayanz, J. (2017). EU-Forest, a high-resolution tree occurrence dataset for Europe. *Scientific Data*, 4(1):160123.

Monnet, A.-C., Cilleros, K., Médail, F., Albassatneh, M. C., Arroyo, J., Bacchetta, G., Bagnoli, F., Barina, Z., Cartereau, M., Casajus, N., Dimopoulos, P., Domina, G., Doxa, A., Escudero, M., Fady, B., Hampe, A., Matevski, V., Misfud, S., Nikolic, T., Pavon, D., Roig, A., Barea, E. S., Spanu, I., Strid, A., Vendramin, G. G., and Leriche, A. (2021). WOODIV, a database of occurrences, functional traits, and phylogenetic data for all Euro-Mediterranean trees. *Scientific Data*, 8(1):89.

Saltré, F., Saint-Amant, R., Gritti, E. S., Brewer, S., Gaucherel, C., Davis, B. A. S., and Chuine, I. (2013). Climate or migration: what limited European beech post-glacial colonization? *Global Ecology and Biogeography*, 22(11):1217–1227.

Valavi, R., Guillera-Arroita, G., Lahoz-Monfort, J. J., and Elith, J. (2022). Predictive performance of presence-only species distribution models: a benchmark study with reproducible code. *Ecological Monographs*, 92(1):e01486.

Van der Meersch, V., Armstrong, E., Mouillot, F., Duputié, A., Davi, H., Saltré, F., and Chuine, I. (2025). Paleorecords reveal biological mechanisms crucial for reliable species range shift projections amid rapid climate change. *Ecology Letters*, 28(2).

Van der Meersch, V. and Chuine, I. (2023). Estimating process-based model parameters from species distribution data using the evolutionary algorithm CMA-ES. *Methods in Ecology and Evolution*, 14(7):1808–1820.